

Calculus: Limits and Derivatives

A limit describes the value a function approaches as input nears a point. The derivative measures instantaneous rate of change. Chain rule: derivative of $f(g(x)) = f'(g(x))$ times $g'(x)$. Product rule: $(uv)' = u'v + uv'$. Integration is the inverse of differentiation. The Fundamental Theorem of Calculus links definite integrals to antiderivatives. Applications include optimization and area under curves.